

# Patient Follow-up Agent - Deployment Evidence

Consolidated evidence pack — eight reports, their raw artifacts, and a re-check of every hash

|                   |                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Prepared for      | Clinic operations and IT decision-makers                                                                                    |
| Purpose           | Deployment acceptance: what was measured, on which revision, and what remains unproven                                      |
| Evidence revision | fbcc6c1ca09a7624fd5f01016e674cacdc4a5287                                                                                    |
| Pack location     | proposal/evidence/                                                                                                          |
| Contents          | 8 numbered reports, 3 raw artifacts, 5 visual artifacts                                                                     |
| Pack provenance   | Reports 1–7 captured by script on 2026-09-25, 09:02–09:47 UTC; report 8 is an operator-supplied screen capture (section 10) |
| Document date     | 26 September 2026                                                                                                           |
| Rendered by       | proposal/scripts/build_deployment_evidence_pdf.py                                                                           |

## 1. How to read this pack

The first seven reports under `proposal/evidence/` were captured independently, each by its own script, on 25 September 2026 between 09:02 and 09:47 UTC. Report 8 was contributed afterwards by the operator as a screen capture, which makes it the one artifact in this pack that no script produced and nobody can re-derive; section 10 states what that costs. This document consolidates the eight: it restates every figure, records the SHA-256 of each artifact as recomputed for this PDF, and keeps the boundary between *measured* and *assumed* explicit.

Three kinds of claim appear in the pack, and they are not interchangeable. It matters which one a figure belongs to when deciding whether the deployment is ready.

| Claim class   | Meaning                                                         | Evidenced by |
|---------------|-----------------------------------------------------------------|--------------|
| Configuration | The deployment is wired to named services with named identities | 2, 3         |
| Behaviour     | The software did what it is specified to do, reproducibly       | 1, 4, 5      |
| Transmission  | A real message left the system and arrived somewhere            | 6, 7, 8      |

Only class three is evidence about the outside world. Evidence 1 and 5 exercise the agent’s real code paths but with the offline printing delivery backend, so they record decisions, not messages sent. Every report’s “what this does *not* claim” section states its own limit, and those limits are reproduced here rather than smoothed over.

### The one-paragraph version

The agent runs its full loop, 811 tests pass on a pinned revision, the dashboard and patient portal serve live data, and the AWS paths are wired to SES and End User Messaging SMS. A real email was transmitted through the agent’s own tool layer, accepted by the provider, counted as delivered in the provider’s telemetry, and then found in the recipient’s **spam** folder; three later reminders were separately observed sitting in that same inbox (report 8). Placement is therefore inconsistent rather than uniformly bad, which does not change the fix — a clinic-owned, authenticated domain, warmed — only the odds of a reminder being seen in the meantime. SES is still in the sandbox and the SMS channel is not subscribed. The deployment is functionally complete and provisionally blocked — the blockers are account provisioning and sender reputation, not code.

## 2. Evidence at a glance

| # | Report                       | What it establishes                                                                                 | Verdict                                 |
|---|------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1 | 01_agent_decision_audit.md   | 42 simulated days, 493 events, 463 decisions, 30 communication attempts over 11 patients            | Recorded                                |
| 2 | 02_aws_messaging_config.md   | Transport pinned to named AWS services, region, sender identity and recipient allow-list            | Configured                              |
| 3 | 03_runtime_and_repository.md | The exact revision, interpreter, dependencies and codebase size every other claim rests on          | Identified                              |
| 4 | 04_test_suite.md             | 811 tests pass, exit code 0, verbatim pytest output retained                                        | Pass                                    |
| 5 | 05_dashboard_runtime.md      | Application starts, serves dashboard and portal, answers its API, runs operator commands            | Running                                 |
| 6 | 06_live_delivery.md          | A real message was handed to the provider through the agent's tool layer and accepted               | Sent                                    |
| 7 | 07_delivery_confirmation.md  | Provider telemetry counted it delivered; the mailbox had filed it as spam                           | <b>Delivered to spam</b>                |
| 8 | 08_patient_inbox_receipt.md  | Three later reminders from the same system seen in the recipient's inbox, per-patient bodies intact | <b>Delivered to inbox</b> (sample data) |

## 3. Evidence 1 — Agent decisions and audit trail

Captured 2026-09-25T09:02:41+00:00.

**Claim.** The agent runs its full Perceive → Decide → Act → Observe loop and writes a durable, queryable audit trail for every decision and every delivery attempt.

**Provenance.** The record is not hand-written. It is the output of the real orchestrator driven by `proposal/scripts/generate_audit_record.py` over the project's sample patients with the clock fixed, so the run does not depend on the wall clock.

| Fact                           | Value                                                               |
|--------------------------------|---------------------------------------------------------------------|
| Snapshot analysed              | proposal/evidence/artifacts/audit_log_snapshot.json (261,297 bytes) |
| Snapshot SHA-256               | 76551f7389296eda3ff1d905ec5ed8c486f98fccd12cff0db80e9cd9bff06644    |
| Events                         | 493 — 463 decisions, 30 communication attempts                      |
| Patients                       | 11 distinct identifiers                                             |
| Simulated period               | 2026-09-21 → 2026-11-01, 42 daily cycles                            |
| Delivery calls handled offline | 37                                                                  |

### Decisions

| Action            | Count | Share |
|-------------------|-------|-------|
| do_nothing        | 430   | 92.9% |
| send_reminder     | 22    | 4.8%  |
| escalate_to_staff | 11    | 2.4%  |

| Urgency at decision time | Count | Case status at decision time | Count |
|--------------------------|-------|------------------------------|-------|
| critical                 | 209   | escalated                    | 282   |
| high                     | 150   | message_sent                 | 98    |
| medium                   | 78    | booked                       | 39    |
| low                      | 26    | declined                     | 32    |
|                          |       | pending                      | 11    |
|                          |       | awaiting_reply               | 1     |

Escalation: 11 actions, 9 distinct patients ever escalated, 2.4% of decisions. The high `do_nothing` share is the intended shape — the agent contacts a patient on a schedule, not every cycle.

Communication attempts

| Channel    | Attempts | Succeeded | Failed | Rate   |
|------------|----------|-----------|--------|--------|
| email      | 12       | 12        | 0      | 100.0% |
| sms        | 9        | 9         | 0      | 100.0% |
| whatsapp   | 7        | 7         | 0      | 100.0% |
| phone_call | 2        | 2         | 0      | 100.0% |
| Total      | 30       | 30        | 0      | 100.0% |

**Reading the 100% rate.** Every attempt succeeded because the backend is the offline printer: it accepts any recipient and cannot fail. The rate describes the offline backend, **not** carrier deliverability, and is not comparable to the fallback-path behaviour in the proposal’s robustness section. Carrier-level delivery is evidenced only by reports 6 and 7.

The Observe path

Five patient replies were injected so the record covers inbound handling, not only outbound decision-making.

| Day        | Patient | Reply                                      | Status before | Status after   |
|------------|---------|--------------------------------------------|---------------|----------------|
| 2026-09-23 | P006    | Yes, I'd like to schedule an appointment   | message_sent  | booked         |
| 2026-09-30 | P007    | No thanks, not needed right now            | message_sent  | declined       |
| 2026-10-07 | P004    | How much will this cost with my insurance? | escalated     | escalated      |
| 2026-10-14 | P003    | Can I come in next week?                   | escalated     | awaiting_reply |
| 2026-10-21 | P011    | Yes please                                 | —             | no_case        |

Final case statuses across the 11 cases: **escalated 9, declined 1, booked 1**.

**The finding that matters here.** Nine of eleven cases end in `escalated`, and by design the agent then stops touching them — clinical ambiguity belongs to a human. But the case is not re-raised if the alert is missed. This is correct behaviour with an operational consequence that belongs in the rollout plan, not a defect to be silently patched.

**Generator guard.** Every constructed channel had to be a `PrintDeliveryBackend`, and the transmitting backend’s `deliver()` was patched to raise. That assertion is what makes this a record of decisions instead of transmission. The sample patients carry invented addresses, so the run must not transmit.

**Data hygiene.** `audit_log.json` is append-only and the automated test suite writes to it, so the live file grows every time the tests run. The figures above come from the frozen snapshot, and re-running the tests will not reproduce identical totals from the live file.

## 4. Evidence 2 — AWS messaging configuration

Captured 2026-09-25T09:30:27+00:00.

**Claim.** The messaging transport is wired to named AWS services in a named region, sends from verified identities, and refuses recipients outside an explicit allow-list.

| Setting                      | Value                                          |
|------------------------------|------------------------------------------------|
| AWS_REGION                   | ap-southeast-1                                 |
| AWS_SES_SOURCE               | brightsmileagent@gmail.com                     |
| AWS_SES_CONFIGURATION_SET    | patient-followup                               |
| AWS_SMS_ORIGINATION_IDENTITY | +6585144321                                    |
| AWS_SMS_ALLOWED_NUMBERS      | +6583536885                                    |
| AWS_EMAIL_ALLOWED_ADDRESSES  | martinchenonly1@gmail.com , rgjh1996@gmail.com |
| MESSAGING_DRY_RUN            | 0                                              |
| AGENT_LIVE_SENDS             | (empty — live sends disabled by default)       |

Read back from the live account: session authenticated as `arn:aws:iam::621236283593:root` ; SES sending enabled `True` ; 24-hour quota `200.0` , max send rate `1.0/s` , sent in the last 24 hours `3.0` . Three verified SES identities ( `brightsmileagent@gmail.com` , `martinchenonly1@gmail.com` , `rgjh1996@gmail.com` ). Verified SMS destination numbers: **not readable** — `SubscriptionRequiredException` from `DescribeVerifiedDestinationNumbers` .

A quota of 200 per 24 hours at 1/s is the SES **sandbox** ceiling, not a production limit. Production access and SMS origination onboarding are prerequisites for a real patient population and are outstanding.

The SMS read failure means this account is not subscribed to End User Messaging SMS in the queried region. The SMS code path is complete, but attempts return `not_subscribed` until the service is provisioned. **Do not describe SMS as available on the strength of this pack.**

Delivery code paths in the captured revision, cited by file and line:

| File                   | Symbol                           | Line |
|------------------------|----------------------------------|------|
| tools/aws_providers.py | client.send_text_message         | 307  |
| tools/aws_providers.py | client.send_email                | 348  |
| tools/messaging.py     | def send_sms                     | 383  |
| tools/messaging.py     | def send_email                   | 413  |
| agent/delivery.py      | def is_configured_for_live_sends | 334  |

## 5. Evidence 3 — Runtime and repository state

Captured 2026-09-25T09:47:34+00:00.

**Claim.** The deployment was validated against a specific, identifiable revision on a specific interpreter with specific library versions, so a reviewer can confirm they are comparing like with like before judging anything else.

**Snapshot scope.** This capture describes commit `fbcc6c1` only. It is **not** a description of the current `master` , which is a later revision. Re-run `proposal/scripts/capture_runtime_evidence.py` to re-capture against a later revision.

| Fact               | Value                                                         |
|--------------------|---------------------------------------------------------------|
| Python             | 3.14.5 (CPython)                                              |
| Executable         | /Users/martinchen/agent_followship_v3/.venv/bin/python        |
| Platform           | macOS-27.0-arm64-arm-64bit-Mach-0                             |
| Branch             | master                                                        |
| Commit             | fbcc6c1ca09a7624fd5f01016e674cacdc4a5287                      |
| Commit subject     | Merge pull request #3 from pioneer6-ai/feature/patient-portal |
| Commit date        | 2026-09-25T13:41:07+08:00                                     |
| Working tree clean | no — 5 files modified (this analysis work only)               |
| Remote             | git@github.com:pioneer6-ai/agent_followship_v2.git            |

Installed dependencies at capture time: boto3 1.35.36, botocore 1.35.99, flask 3.0.0, openpyxl 3.1.2, certifi 2026.7.22, **anthropic not installed**.

Codebase size — 78 files, 32,073 lines:

| Module                                           | Files | Lines  |
|--------------------------------------------------|-------|--------|
| tests/                                           | 30    | 12,119 |
| tools/ (messaging, AWS, LLM, transport)          | 13    | 6,118  |
| agent/ (orchestrator, decision, delivery, audit) | 9     | 4,635  |
| web/ (dashboard, portal, APIs)                   | 5     | 3,477  |
| core/ (models, clock, policy, scheduling)        | 11    | 1,978  |
| scheduling/                                      | 4     | 1,321  |
| utils/                                           | 2     | 1,022  |
| root( app , demo )                               | 2     | 1,115  |
| scripts/                                         | 2     | 288    |
| Total                                            | 78    | 32,073 |

**Path caveat, stated by the pack itself.** The interpreter recorded here lives in the sibling clone `agent_followship_v3` , because that is where the virtualenv that produced the capture sits. The clone this PDF ships in ( `agent_followship_v2` ) has no `.venv` , and its installed dependency set is the one listed in section 12 rather than the list above. The evidence is valid for the recorded revision and the recorded interpreter; it is not a claim about the virtualenv in this directory.

## 6. Evidence 4 — Automated test suite

Captured 2026-09-25T09:47:34+00:00.

**Claim.** The revision identified in evidence 3 passes its full automated suite on the recorded interpreter. The suite is the executable form of the behaviour contracts: tool schemas, error taxonomy, provider selection, agent loop control flow, delivery gating and the web API.

**Snapshot scope.** The result below ( `811 passed` ) is the suite as it stood at commit `fbcc6c1` only. It is **not** the current count on `master` .

| Fact                               | Value                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------|
| Status                             | <b>PASS</b> (exit code 0)                                                              |
| Command                            | <code>/Users/martinchen/agent_followship_v3/.venv/bin/python -m pytest tests -q</code> |
| Passed / Failed / Errors / Skipped | 811 / 0 / 0 / 0                                                                        |
| Warnings                           | 16                                                                                     |
| Duration                           | 14.43 s                                                                                |
| Verbatim output                    | <code>proposal/evidence/artifacts/pytest_output.txt</code>                             |
| Summary line                       | <code>811 passed, 16 warnings in 14.43s</code>                                         |

The inventory counts test **functions** — 681 across 29 modules, 11,915 lines of test code. pytest reports 811 passing **cases** because several are parameterised and expand at collection time. Both numbers describe one suite.

The heaviest modules by test count: `test_llm_providers.py` (79), `test_aws_tools.py` (73), `test_hospital_setup.py` (64), `test_agent_delivery.py` (50), `test_providers.py` (48), `test_agent_loop.py` (39), `test_patient_portal.py` (31), `test_tool_contract.py` (29).

**Known limitation, found while capturing this evidence.** The suite exercises the real orchestrator and therefore its real audit logger, which appends to `audit_log.json` in the working directory. Running the tests mutates the audit record. Evidence 1 is computed from a frozen snapshot for exactly this reason, and isolating the audit path during tests is a prerequisite for production sign-off.

## 7. Evidence 5 — Running dashboard and patient portal

Captured 2026-09-25T09:06:57+00:00.

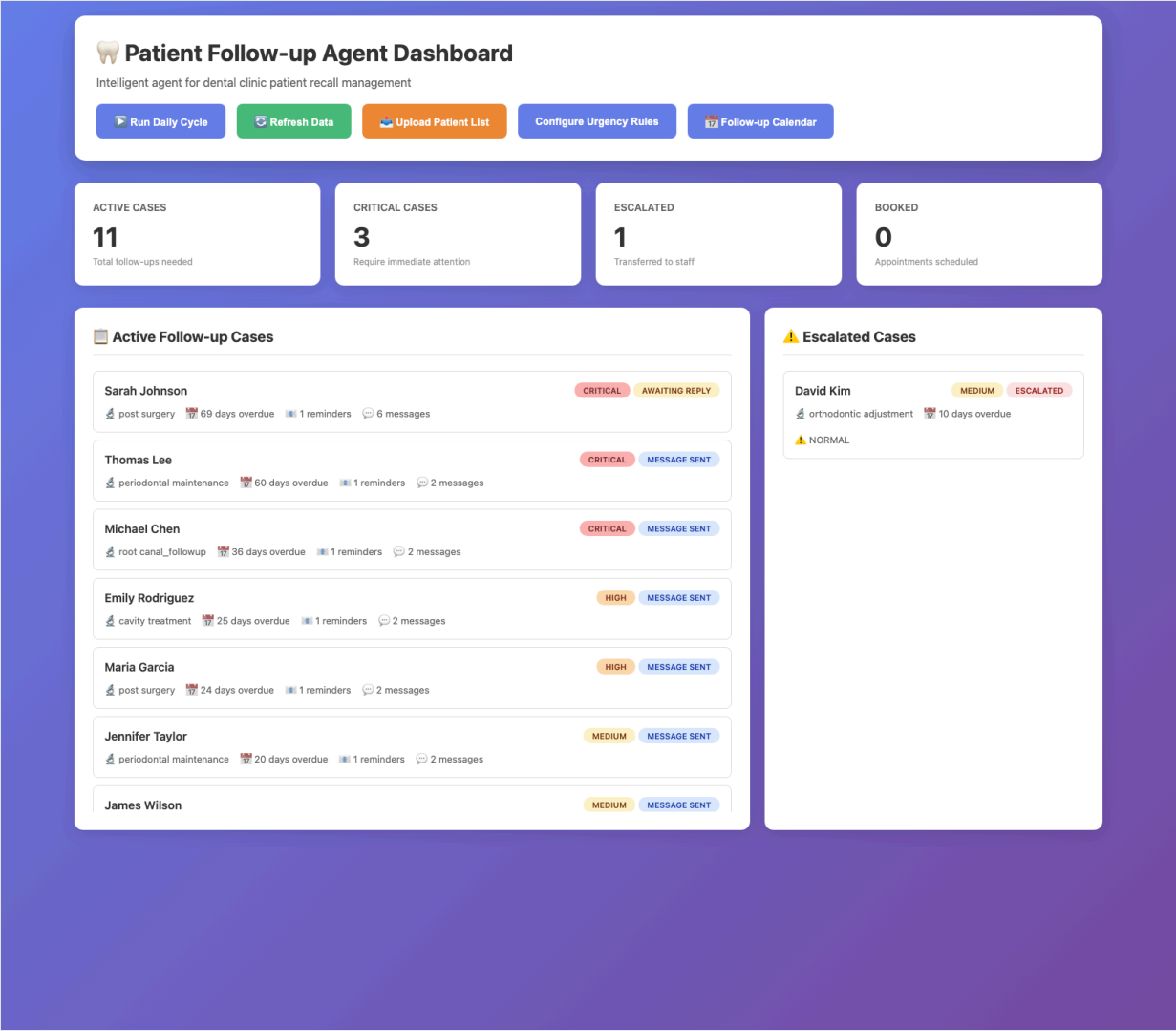
**Claim.** The deployed application starts, serves its staff dashboard and its patient portal, answers its JSON API and accepts operator commands. The screenshots are the UI as a browser receives it; the JSON files are the unedited API responses the dashboard consumes while rendering.

| Fact                      | Value                                                                                |
|---------------------------|--------------------------------------------------------------------------------------|
| Server                    | <code>http://127.0.0.1:54301</code> (werkzeug, bound to localhost)                   |
| Dashboard screenshot      | 475,711 bytes, viewport 1600×1400                                                    |
| Patient portal screenshot | 736,923 bytes                                                                        |
| Portal URL                | <code>/patient/QITt-c7mJE6WhJtNDUIjdVGqpQPPrnJs</code> (token scoped to one patient) |

The page was not captured idle. Before the screenshot, the same endpoints the dashboard’s buttons call were invoked, so the board shows cases this process produced:

- `POST /api/run-cycle` → `success=True` , `cases_processed=11`
- `POST /api/simulate-reply` P001 *“Yes, I’d like to schedule an appointment”* → `message_sent` → `awaiting_reply`
- `POST /api/simulate-reply` P007 *“No thanks, not needed right now”* → `message_sent` → `declined`
- `POST /api/simulate-reply` P004 *“How much will this cost with my insurance?”* → `message_sent` → `escalated`

The running board, 11 cases in five states, urgency critical 3 / high 2 / medium 3 / low 3:



The operator dashboard, populated by the capture's own agent run

The patient portal, rendered from a token minted by the running application rather than mocked up:

Hi Sarah Johnson 🍌

Pick a time that works for you below.

awaiting reply

Available appointment times

9月28日星期一

9:00 AM

9:30 AM

10:00 AM

10:30 AM

11:00 AM

11:30 AM

1:00 PM

1:30 PM

2:00 PM

2:30 PM

3:00 PM

3:30 PM

4:00 PM

4:30 PM

9月29日星期二

9:00 AM

9:30 AM

10:00 AM

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11:30 AM

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11:30 AM

1:00 PM

1:30 PM

2:00 PM

2:30 PM

3:00 PM

3:30 PM

4:00 PM

4:30 PM

None of these times work — remind me next week

Questions? Ask our assistant

Ask about your appointment or general dental care...

Send

The patient portal, served by the same process

GET /api/status :

```
{
  "audit_summary": {
    "successful_bookings": 0,
    "total_communications": 17,
    "total_decisions": 12,
    "total_log_entries": 29
  },
  "cases_by_status": {
    "awaiting_reply": 1, "booked": 0, "declined": 1, "escalated": 1,
    "message_sent": 8, "opted_out": 0, "pending": 0,
    "pending_future_availability": 0
  },
  "cases_by_urgency": { "critical": 3, "high": 2, "low": 3, "medium": 3 },
  "escalated_cases": 1,
  "last_updated": "2026-09-25T17:06:55.280205",
  "total_active_cases": 11,
  "total_patients": 12
}
```

GET /api/cases returned 11 cases; first entry:

```
{
  "conversation_length": 6,
  "days_overdue": 69,
  "last_contacted": "2026-09-25",
  "last_visit": "2026-06-27",
  "patient_id": "P001",
  "patient_name": "Sarah Johnson",
  "preferred_channel": "sms",
  "reminder_count": 1,
  "status": "awaiting_reply",
  "treatment_type": "post_surgery",
  "urgency": "critical"
}
```

The board is populated with the project's built-in sample data, not real patients. It demonstrates the operator experience and the operating surface; it is not a record of live patient activity.

**Audit isolation.** This run wrote its audit log to a temporary directory ( `followup-dashboard-*/audit_log.json` ) and discarded it, so exercising the agent here cannot alter the audit trail evidence 1 is computed from. The two captures were deliberately separated for this reason.

## 8. Evidence 6 — Live end-to-end delivery

Captured 2026-09-25T09:30:36+00:00, revision `fbcc6c1ca09a7624fd5f01016e674cacdc4a5287` .

**Claim.** A message was handed to the real provider through the agent's own tool layer and the provider answered. This is one of only two artifacts in the pack that evidences transmission rather than behaviour, which is why it refuses to run without explicit approval.

| Fact                   | Value                                                                         |
|------------------------|-------------------------------------------------------------------------------|
| Tool                   | <code>send_email</code>                                                       |
| Channel                | <code>email</code>                                                            |
| Recipient              | <code>martinchenonly1@gmail.com</code>                                        |
| Sender identity        | <code>brightsmileagent@gmail.com</code>                                       |
| Reason recorded        | <code>"pilot acceptance check: verifying the delivery path end to end"</code> |
| Latency                | <code>228 ms</code>                                                           |
| <code>simulated</code> | <code>false</code>                                                            |

Provider response:

```
{
  "status": "sent",
  "channel": "email",
  "recipient": "martinchenonly1@gmail.com",
  "data": { "reason": "pilot acceptance check: verifying the delivery path end to end" },
  "message_id": "010e01a0d7e70489-3cad714-dbd2a-4219-9174-efc749f3ae5f-000000"
}
```

The toolkit's own audit trail for the same run records the tool, channel, recipient, status, message id, null error fields, latency, reason and body preview — the same fields production relies on, emitted by the same code path.

**Acceptance is not delivery.** The provider returning `sent` means SES accepted the message for transmission. For email, the authoritative record of arrival is the `Delivery` event emitted by the `patient-followup` configuration set, which is what evidence 7 reads.

## 9. Evidence 7 — Delivery confirmation

Captured 2026-09-25T09:47:34+00:00.

**Claim.** What happened to the message after the provider accepted it, from two independent vantage points: the provider's own delivery telemetry and the recipient's mailbox. These are different claims and are kept separate on purpose.

Message under test: id 010e01a0d7e70489-3cad714-db2a-4219-9174-efc749f3ae5f-000000 , recipient martinchenonly1@gmail.com , sent 2026-09-25T09:30:36+00:00.

Check 1 — provider delivery telemetry

CloudWatch AWS/SES for configuration set patient-followup , namespace queried in ap-southeast-1 :

| Metric    | Sum |
|-----------|-----|
| Delivery  | 6   |
| Bounce    | 0   |
| Complaint | 0   |

No bounce and no complaint across the window is a genuine positive: the sending identity is not being rejected or abused-flagged.

Check 2 — recipient mailbox

Read-only IMAP search of imap.gmail.com for subject fragment “delivery confirmation”:

| Folder   | Messages found |
|----------|----------------|
| INBOX    | 0              |
| spam     | 5              |
| all_mail | 0              |

The message was found in: spam.

This is the most consequential operational finding in the pack. The mailbox accepted the message and filed it as spam. A reminder that lands in a spam folder has not reminded anyone, and no amount of agent correctness compensates for it. This is a sending-identity and reputation problem, not a software problem: it is fixed by moving to a clinic-owned domain with SPF, DKIM and DMARC configured, and by warming that domain.

Observed headers (three of the five messages found):

Subject: Patient Follow-up Agent - delivery confirmation  
From: brightsmileagent@gmail.com  
Date: Fri, 25 Sep 2026 09:30:36 +0000  
Message-ID: <010e01a0d7e70489-3cad714-db2a-4219-9174-efc749f3ae5f-000000@ap-southeast-1.amazonaws.com>

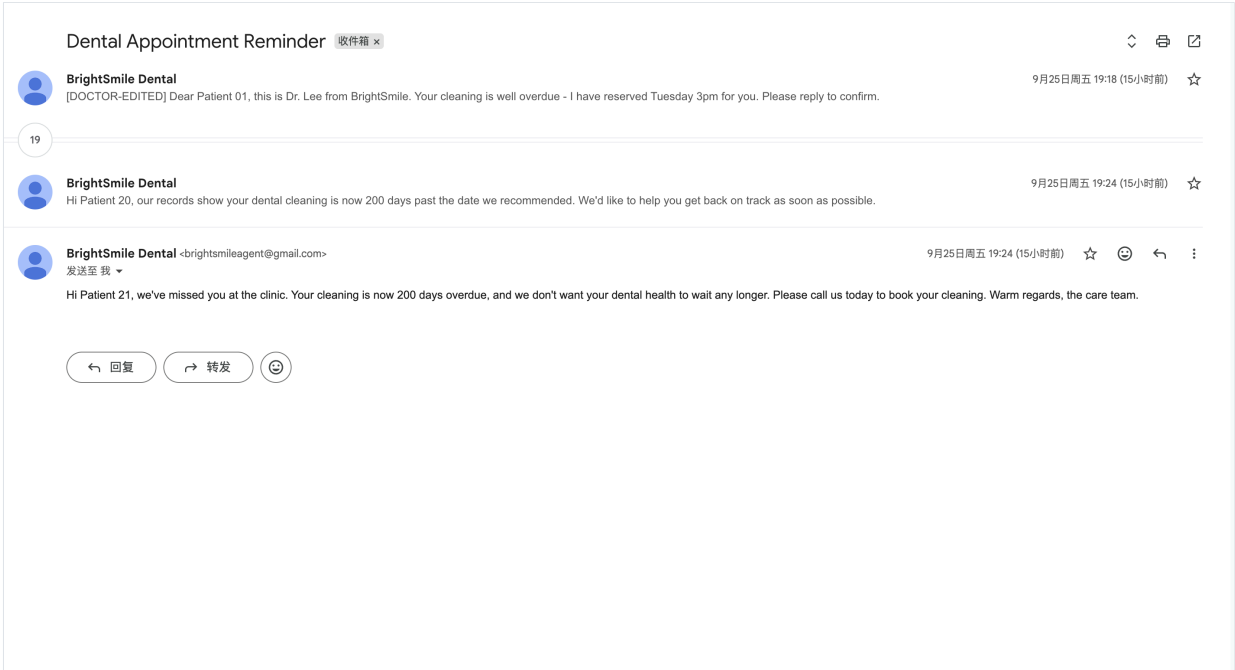
Subject: Patient Follow-up Agent - delivery confirmation  
From: brightsmileagent@gmail.com  
Date: Fri, 25 Sep 2026 09:30:27 +0000  
Message-ID: <010e01a0d7e6e250-35534990-542e-4545-8385-16806cc2153c-000000@ap-southeast-1.amazonaws.com>

Subject: Patient Follow-up Agent - delivery confirmation  
From: brightsmileagent@gmail.com  
Date: Fri, 25 Sep 2026 09:30:11 +0000  
Message-ID: <010e01a0d7e6a32e-de7255b4-a647-42f4-8e67-7c915e659186-000000@ap-southeast-1.amazonaws.com>

10. Evidence 8 — Patient inbox receipt

Contributed 2026-09-26T10:36+08:00 by the operator as a screen capture.

**Claim.** Reminders written by the agent and sent through the configured SES identity reached the recipient’s **inbox**, rendered as ordinary mail from “BrightSmile Dental”, with the per-patient generated wording intact. This is a third vantage point on delivery alongside the provider telemetry and the IMAP search in evidence 7, and it disagrees with evidence 7 on where the mail was filed.



Three reminders from BrightSmile Dental sitting in the recipient’s inbox

|               |                                                                                  |
|---------------|----------------------------------------------------------------------------------|
| File          | proposal/evidence/screenshots/patient_inbox_received.png                         |
| Size          | 365,876 bytes                                                                    |
| Dimensions    | 2,840 x 1,538, 8-bit RGBA PNG                                                    |
| SHA-256       | 6302c9b8941f048ede4c527671513d48be1188bfa469c5c5e0e4bd94ea70a2c9                 |
| Thread header | 9月25日周五 19:18–19:24（15小时前） — Friday 25 September, 19:18 to 19:24, “15 hours ago” |

Three messages in one thread, every sender line reading “BrightSmile Dental”. The first two are collapsed to their opening line, the third is expanded:

| Order | Opening line as displayed                                                                                                                                                                                               | Addressed to |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1     | [DOCTOR-EDITED] Dear Patient 01, this is Dr. Lee from BrightSmile. Your cleaning is well overdue – I have reserved Tuesday 3pm for you. Please reply to confirm.                                                        | Patient 01   |
| 2     | Hi Patient 20, our records show your dental cleaning is now 200 days past the date we recommended. We'd like to help you get back on track as soon as possible.                                                         | Patient 20   |
| 3     | Hi Patient 21, we've missed you at the clinic. Your cleaning is now 200 days overdue, and we don't want your dental health to wait any longer. Please call us today to book your cleaning. Warm regards, the care team. | Patient 21   |

The three bodies differ in greeting, phrasing and closing. These are not one template with a substituted name, and none of those sentences appears in any template in the repository.

What corroborates it

The proposal already documents this batch. Section 9.4 records the trial: a model configured for decisioning and message authoring, run against a cohort of 21 patients all scored critical, a staff edit honoured, then “Confirm Selected” reporting 21 sent, 0 failed, queue empty. Section 10 is the recipient-side view of that same event — the console and the mailbox, one batch.

The run’s audit log puts the same messages in the same window. The log for that deployment (the sibling clone’s append-only `audit_log.json`, which section 9.4 explains is deliberately kept out of the pack) contains:

| Timestamp (+08:00)  | Patient | Channel | Success | Preview                                                                                                 |
|---------------------|---------|---------|---------|---------------------------------------------------------------------------------------------------------|
| 2026-09-25T19:18:10 | CRIT001 | email   | true    | [DOCTOR-EDITED] Dear Patient 01, this is Dr. Lee from BrightSmile. Your cleaning is well overdue – I... |
| 2026-09-25T19:23:03 | CRIT001 | email   | true    | [DOCTOR-EDITED] Dear Patient 01, this is Dr. Lee from BrightSmile. Your cleaning is well overdue – I... |
| 2026-09-25T19:24:06 | CRIT020 | email   | true    | Hi Patient 20, our records show your dental cleaning is now 200 days past the date we reco...           |
| 2026-09-25T19:24:08 | CRIT021 | email   | true    | Hi Patient 21, we've missed you at the clinic. Your cleaning is now 200 days overdue, and...            |

The first row is a single send of the edited draft; the batch is 21 sends to CRIT001 – CRIT021 between 19:23:03 and 19:24:08, all channel=email, all direction=outbound, all success=true, which is section 9.4’s 21 sent, 0 failed. Two details identify the log as that run rather than a similar one: it carries exactly the 11 llm-error and 1 llm-guardrail entries section 9.4 quotes, and its Decided by: llm count has grown from the 944 section 9.4 records to 1,600 — an append-only record still being written to, which is the reason it stays out of the pack. A reviewer can therefore check this section’s arithmetic against test\_data/demo\_21\_critical\_patients.csv and the proposal, but not against a file this pack ships, and report 8 says so.

Two further cross-checks, both recomputable from files in this repository:

- CRIT001, CRIT020 and CRIT021 are the ID column of test\_data/demo\_21\_critical\_patients.csv: 21 rows named Patient 01 to Patient 21, every one carrying martinchenonly1@gmail.com — one of the two addresses in the AWS\_EMAIL\_ALLOWED\_ADDRESSES allow-list of evidence 2. That is why three patients share one mailbox and one thread.
- The “200 days” in the body is not decorative. Those rows record a last visit of 2026-02-23 and a recall window of 14 days, so the cleaning fell due on 2026-03-09; on 25 September 2026 that is exactly 200 days overdue.

What it does not prove

This is the one artifact in the pack that no capture script produced. No one can re-run it, and the hash above pins the bytes this pack ships, not the provenance of the screen it records.

- **That the doctor-edited marker is generated.** [DOCTOR-EDITED] appears nowhere in the repository; a case-insensitive search of the revision returns no source file. It is text the operator typed into the draft body, carried through the draft-edit path ( update\_pending\_send / edited\_by in agent/orchestrator.py, reached from web/app.py ), not a label the software emits.
- **That any real patient was contacted.** The recipients are the sample list.
- **That the message arrived for the other 18 patients, or anywhere else.** One mailbox, one provider, three of 21 messages shown: an existence proof of inbox placement, not a placement rate.
- **That a delivered reminder is a read reminder.** An inbox is not an open, a reply or a booking.
- **That evidence 7 was wrong.** Evidence 7’s read-only IMAP search found its own subject fragment in spam. Both observations hold: different messages, different times, and placement for a new sending identity on a free mailbox provider is exactly the sort of thing that varies. The domain and reputation work stays necessary; this artifact does not reduce it.
- **That the capture sits inside the pack’s own capture window.** 19:18–19:24 +08:00 is 11:18–11:24 UTC, whereas every scripted capture ran between 09:02 and 09:47 UTC that day. It is a later event observed after the fact, not a re-render.

## 11. Artifact integrity — hashes as recomputed for this PDF

The pack's value rests on artifacts being the same bytes the reports were written from. Every artifact was re-hashed for this document. Full SHA-256, with the size recorded in `00_index.md` beside the size on disk. One row is marked hand-contributed: its hash pins the bytes this pack ships, but no capture script produced it, so there is nothing for a reviewer to re-run.

| Artifact                                            | Size on disk | SHA-256                                                                        | Matches index                      |
|-----------------------------------------------------|--------------|--------------------------------------------------------------------------------|------------------------------------|
| <code>00_index.md</code>                            | 5,728        | <code>d48bdd21b125ba4cdd61a44fdcb46a2a73cb621851b01f670d60bbdded6dd2c8d</code> | n/a (is the index)                 |
| <code>01_agent_decision_audit.md</code>             | 5,155        | <code>1b9c73463a6718c18e0373640ef64cc77a918d383295a1c5b460fbca3f573f41</code>  | yes                                |
| <code>02_aws_messaging_config.md</code>             | 2,302        | <code>2dab4ce38689760c3ef5184f86c2b5666abf9173876491f6228c863d7ee43f01</code>  | yes                                |
| <code>03_runtime_and_repository.md</code>           | 1,788        | <code>f7de4397ab358fde7f00f17e735a9540e258a05751843b4f1335b405e23de4f5</code>  | yes (was stale — see below)        |
| <code>04_test_suite.md</code>                       | 3,228        | <code>a53602b82f2e15354317ec7298f7427947a6d3fcc5dd5911342f46e713308af6</code>  | yes (was stale — see below)        |
| <code>05_dashboard_runtime.md</code>                | 3,368        | <code>9a800d1e377ede06f2f538091a763c46822b3e98a4f1b37b3b8daadfbe3d9a52</code>  | yes                                |
| <code>06_live_delivery.md</code>                    | 2,092        | <code>ba29e0695c1cc155a25e605f7cd89d0d6dc77191d4c38604e9d5f108e24e4553</code>  | yes                                |
| <code>07_delivery_confirmation.md</code>            | 2,589        | <code>5f9b34db0ca43398bee5596d990de0985a0c86153783a74421c6f6e8b277d629</code>  | yes                                |
| <code>08_patient_inbox_receipt.md</code>            | 8,863        | <code>6ff287cbeead04c5491f5004822d24409f8080bfc14b145fdd0e046d3e47d0a5</code>  | yes                                |
| <code>artifacts/audit_log_snapshot.json</code>      | 261,297      | <code>76551f7389296eda3ff1d905ec5ed8c486f98fccd12cff0db80e9cd9bffb06644</code> | yes                                |
| <code>artifacts/audit_log.generation.json</code>    | 4,782        | <code>42da6d67abc3fdad5415daab3e377b2b44c9c3531acadb79079b20df89eb59243</code> | yes                                |
| <code>artifacts/pytest_output.txt</code>            | 2,216        | <code>1da74e80729d72a81aeb2561d0504b1ee665df426d930122ac12dfb41feefcf6</code>  | yes                                |
| <code>screenshots/dashboard_cases_api.json</code>   | 3,824        | <code>393bcb9f050eb10236fdde27ce9dcf8658ef16ef96df26e29f1a0fbc3a6e3da</code>   | yes                                |
| <code>screenshots/dashboard_overview.png</code>     | 475,711      | <code>3c4e06f485f6bf9d95226442f8c71957674596d287a2ea7612e50c603ffc5e7a</code>  | yes                                |
| <code>screenshots/dashboard_status_api.json</code>  | 569          | <code>acb61130deebd2a46f6c3d05620cecf205514c4dcbb67fc315511d9448e3534a</code>  | yes                                |
| <code>screenshots/patient_inbox_received.png</code> | 365,876      | <code>6302c9b8941f048ede4c527671513d48be1188bfa469c5c5e0e4bd94ea70a2c9</code>  | yes (hand-contributed; section 10) |
| <code>screenshots/patient_portal.png</code>         | 736,923      | <code>b919902b944a98d3cc730abafc2248ea60f991bce62c82bbdac700dfcd486b20</code>  | yes                                |

### The two stale index entries — and the one artifact no script produced

`00_index.md` records hashes and sizes for every report. In the revision this pack was first built from, entries 3 and 4 did not match their files. That is exactly the failure mode a hash exists to catch, so it was reported rather than quietly fixed, and it is recorded here for the same reason — as a note on how the pack was maintained, not as an open item. It is now closed:

- Commit `0735515` (“docs: point project references at `agent_followship_v2` and stamp evidence snapshots”) added a *snapshot scope* note to `03_...` `md` and `04_...` `md` and repointed the recorded interpreter from this clone’s absent `.venv` to `agent_followship_v3/.venv`. The commit body states plainly that verbatim captured output was deliberately left untouched.
- The commit did **not** re-run `proposal/scripts/capture_evidence_index.py`, so the index carried the pre-edit sizes and hashes for those two files.
- The *numbers* in reports 3 and 4 were never in question: they still trace to their JSON sidecars and to `artifacts/pytest_output.txt`. Only the index’s byte-for-byte attestation of the two prose files was wrong.
- Re-running the index capture against the pack as it stands in this revision resolves it: all eight reports and every supporting artifact now match their indexed size and hash. The index also gained the hand-contributed entry for report 8’s image, and its preamble now states that an artifact may be contributed by hand rather than captured, because leaving the old wording in place would have claimed a script for that image that does not exist.

Two consequences of report 8 remain, and they are properties of the artifact rather than of the index. Its hash is verifiable; its provenance is not — nothing in the pack can show that the image is a screen of the mailbox it says it is. And the index’s attestation for it is only as good as the file the operator supplied.

Reviewing this pack, in order

- 1. Confirm the revision on the first page matches the build under review.
- 2. Check `artifacts/pytest_output.txt` against the hash in section 11 — that is the one artifact carrying verbatim program output.
- 3. Re-derive the evidence 1 counts from `audit_log_snapshot.json` rather than reading the tables in section 3.
- 4. Read section 13 before accepting the “reading the 100% rate” qualification in section 3 as a caveat rather than a defect.
- 5. Treat section 10 as the weakest link in the chain of custody: it is the only claim here whose source process a reviewer cannot repeat.

12. Independent re-run on the current revision

The reports above are pinned to `fbcc6c1`. The working tree that first shipped this PDF is at `0735515` (`master`, 2026-09-26), one commit later, and the pack’s own snapshot-scope notes ask a reader to re-capture rather than extrapolate. The revision that ships this document adds report 8 and the index regeneration — documentation and one image, no application code — so the re-run below still describes the current revision. The suite was re-run here, on this clone, to see whether the pass result survives the later revision and a different interpreter. It does:

| Fact            | Evidence 4 ( <code>fbcc6c1</code> )                | This re-run ( <code>0735515</code> )                      |
|-----------------|----------------------------------------------------|-----------------------------------------------------------|
| Passed          | 811                                                | 934                                                       |
| Skipped         | 0                                                  | 1                                                         |
| Failed / Errors | 0 / 0                                              | 0 / 0                                                     |
| Interpreter     | CPython 3.14.5 (sibling clone <code>.venv</code> ) | CPython 3.13.15 (conda env <code>agent_hackathon</code> ) |
| Duration        | 14.43 s                                            | 10.74 s                                                   |

The suite grew by 123 cases between the two revisions, which follows from the commits in between (the LLM message-authoring port and the docs work). The one skip is deliberate and reported by `pytest` as a skip, not a failure.

Two honest qualifications. First, this re-run used the conda environment `agent_hackathon`, whose dependency set is the `requirements.txt` pin list — `flask 3.0.0`, `werkzeug 3.0.1`, `openai 1.51.0`, `anthropic 1.8.0`, `boto3 1.35.36`, `botocore 1.35.99`, `openpyxl 3.1.2`, `pytest 7.4.3`. It is *not* the interpreter that produced evidence 4. Second, running the suite exercises the real audit logger, so this run appended to the working tree’s `audit_log.json` — the same limitation evidence 4 records. It cannot affect evidence 1, which is computed from the frozen snapshot.

Nothing here is a substitute for evidence 4: it is a corroboration at a later revision, on a different interpreter, and it should be read that way.

13. What the pack proves, and what it does not

Proven.

- The agent runs its full loop and produces a durable audit trail: 493 events over a 42-day simulated period (evidence 1).
- The revision, interpreter, dependency pins and codebase size are identified precisely enough that another party can reproduce the comparison (evidence 3).
- The full suite passes at that revision, exit code 0, with verbatim output retained (evidence 4), and it still passes one revision later on a different interpreter (section 12).
- The application starts, serves the dashboard, the patient portal and the JSON API, and executes operator commands against its own running process (evidence 5).
- SES accepted a message produced by the agent’s own tool layer and returned a message id (evidence 6); the provider’s telemetry counted deliveries with zero bounces and zero complaints (evidence 7).
- Reminders from this system can land in an inbox and render as ordinary mail from the configured sender, with per-patient generated wording and a figure that recomputes from the sample data (evidence 8). Existence, not rate.

Not proven, and not to be inferred from the above.

- **Inbox delivery as a property.** The delivery-confirmation message was filed as spam (evidence 7) while three later reminders were observed in the inbox (evidence 8), from one mailbox and one provider. Placement is inconsistent, so no reminder can be assumed to have been seen. Until sending moves to a clinic-owned domain with SPF, DKIM and DMARC, this stays a coin toss that happens to have landed well twice.
- **Real patient transmission.** Evidence 1’s 100% success rate is a property of the offline printing backend; evidence 5 runs on built-in sample data.
- **SMS.** The channel is code-complete but the account is not subscribed; attempts return `not_subscribed` (evidence 2).
- **Production throughput.** The recorded quota is the SES sandbox ceiling of 200 per 24 h at 1/s (evidence 2).

- **The model-backed decision path as a pinned artifact.** The pack's decision counts come from the rule engine because the evidence harness pins it, not because a model was unavailable. The deployment is wired for a model; the pack contains no hash-identified artifact for that path.
- **Alert re-raise after escalation.** Nine of eleven cases end escalated and are then left alone by design (evidence 1). That is correct behaviour with an operational gap behind it if a staff alert is missed.
- **Readership.** Evidence 8 shows three reminders in an inbox. It does not show an open, a reply or a booking, and no patient behaviour is evidenced anywhere in the pack.
- **Provenance of report 8's image.** Its bytes are pinned by hash; that it is a screen of the mailbox it depicts is the operator's word, and the report says so itself (section 11).

## 14. Open items this pack hands to provisioning

None of these are development tasks, and each one is the difference between a demonstrated system and a usable one:

| # | Item                                                                                                                                 | Evidence      | Blocking for                                                                 |
|---|--------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------|
| 1 | Move SES out of the sandbox: request production access                                                                               | 2             | Any population larger than verified test recipients                          |
| 2 | Send from a clinic-owned domain with SPF, DKIM, DMARC; warm it                                                                       | 7, 8          | Reminders reaching an inbox consistently rather than sometimes               |
| 3 | Subscribe End User Messaging SMS and onboard an origination identity                                                                 | 2             | The SMS channel, and therefore the fallback path                             |
| 4 | Isolate the audit path during tests                                                                                                  | 4             | Production sign-off; currently tests mutate the live audit record            |
| 5 | Decide the escalation re-raise policy — re-alert after N hours, or accept the gap                                                    | 1             | Whether a missed staff alert has a safety net                                |
| 6 | Decide how a hand-contributed artifact is attested — accept the operator's provenance, or require a capture that a script can repeat | 10, 11        | Whether any future screenshot can be treated as more than an existence proof |
| 7 | Pin the model-backed decision path to its own hash-identified artifact                                                               | proposal §7.2 | Any claim about model-driven decisions                                       |

## 15. Reproducing this pack

Every report is regenerated by one command, from the project root, in an environment with the project's dependencies installed:

```
python proposal/scripts/generate_audit_record.py           # regenerate audit_log.json
python proposal/scripts/capture_audit_evidence.py          # evidence 1
python proposal/scripts/capture_aws_config_evidence.py     # evidence 2
python proposal/scripts/capture_runtime_evidence.py        # evidence 3
python proposal/scripts/capture_tests_evidence.py          # evidence 4
python proposal/scripts/capture_dashboard_screenshot.py    # evidence 5
python proposal/scripts/capture_live_send.py --channel email --approve # evidence 6
python proposal/scripts/capture_delivery_confirmation.py    # evidence 7
python proposal/scripts/capture_evidence_index.py          # index
python proposal/scripts/build_deployment_evidence_pdf.py   # this document
```

Evidence 6 transmits to a real mailbox and therefore refuses to run without `--approve`; every other capture script is read-only with respect to the outside world. Evidence 1 depends on the generator in the first line, because the audit log is appended to rather than recomputed.

Report 8 has no command in that list and cannot have one: its image is a screen capture contributed by the operator, so there is nothing to re-run. The pack's hand-contributed artifacts are declared in `proposal/scripts/capture_evidence_index.py` ( `HAND_CONTRIBUTED` ) precisely so the generated index keeps saying so.

Three prerequisites apply. Evidences 2, 6 and 7 read live AWS state, so the shell must hold valid credentials for the account. Evidences 5 and 7 drive a local browser and an IMAP client respectively (headless Chrome, and the mailbox credentials from `SMTP_USERNAME` / `SMTP_PASSWORD` in `.env`). Evidence 5 should be re-captured in a scratch directory so its agent run cannot append to the audit trail evidence 1 is derived from.

Each capture writes both a human-readable report and the machine-readable facts behind it, plus the raw artifact where one exists, so a reviewer can recompute any figure in this document instead of trusting it. `00_index.md` records the SHA-256 of every artifact at the moment the index was generated.

This PDF itself is generated, not assembled by hand: it is `proposal/deployment_evidence.md` rendered through the same pandoc → headless Chrome pipeline as the proposal, with the shared print stylesheet in `proposal/assets/proposal.css`. Re-running the build reproduces it byte for byte from the markdown, and the script verifies the result rather than assuming success — it checks the PDF magic bytes, rejects a suspiciously small file, and reports the page count.